

Database Technologies – Assignment 10

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = $2*(L+W)$ and Area = $L*W$).

```
KD1_Rosh_98811> DELIMITER $$
KD1_Rosh_98811> CREATE PROCEDURE sp_rectangle()
-> BEGIN
-> DECLARE l DOUBLE DEFAULT 10;
-> DECLARE w DOUBLE DEFAULT 5;
-> DECLARE v_perimeter DOUBLE DEFAULT 2*(l+w);
-> DECLARE v_area DOUBLE DEFAULT l*w;
-> SELECT v_perimeter AS perimeter, v_area AS area;
-> END;
-> $$
ERROR 1304 (42000): PROCEDURE sp_rectangle already exists
KD1_Rosh_98811> DELIMITER ;
KD1_Rosh_98811> CALL sp_rectangle();
+-----+-----+
| perimeter | area |
+-----+-----+
|        30 |    50 |
+-----+-----+
1 row in set (0.006 sec)

Query OK, 0 rows affected (0.023 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
KD1_Rosh_98811> DROP TABLE IF EXISTS temp;
Query OK, 0 rows affected (0.189 sec)

KD1_Rosh_98811> CREATE TABLE temp(num INT, square_num INT, cube_num INT);
Query OK, 0 rows affected (0.370 sec)

KD1_Rosh_98811> DELIMITER $$
KD1_Rosh_98811> CREATE PROCEDURE sp_num_square_cube()
-> BEGIN
-> DECLARE num INT DEFAULT 5;
-> INSERT INTO temp VALUES(num, num*num, num*num*num);
-> END;
-> $$
ERROR 1304 (42000): PROCEDURE sp_num_square_cube already exists
KD1_Rosh_98811> DELIMITER ;
KD1_Rosh_98811> CALL sp_num_square_cube();
Query OK, 1 row affected (0.036 sec)

KD1_Rosh_98811> SELECT * FROM temp;
+-----+-----+-----+
| num | square_num | cube_num |
+-----+-----+-----+
|    5 |          25 |        125 |
+-----+-----+-----+
1 row in set (0.004 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F-32) * 5/9$ $F = 9/5 * C + 32$

```
KD1_Rosh_98811> DELIMITER $$
KD1_Rosh_98811> CREATE PROCEDURE sp_temperature(p_value DOUBLE, p_choice INT)
-> BEGIN
-> IF p_choice=1 THEN
-> SELECT (p_value-32)*5/9 AS Celsius;
-> ELSE
-> SELECT 9*p_value/5+32 AS Fahrenheit;
-> END IF;
-> END;
-> $$
ERROR 1304 (42000): PROCEDURE sp_temperature already exists
KD1_Rosh_98811> DELIMITER ;
KD1_Rosh_98811> CALL sp_temperature(100,1);
+-----+
| Celsius |
+-----+
| 37.77777777777778 |
+-----+
1 row in set (0.004 sec)

Query OK, 0 rows affected (0.022 sec)

KD1_Rosh_98811> CALL sp_temperature(37,2);
+-----+
| Fahrenheit |
+-----+
|          98.6 |
+-----+
1 row in set (0.003 sec)

Query OK, 0 rows affected (0.012 sec)
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
KD1_Rosh_98811> DELIMITER $$
KD1_Rosh_98811> CREATE PROCEDURE sp_convert_inches(p_inches INT)
-> BEGIN
-> DECLARE yards INT;
-> DECLARE feet INT;
-> DECLARE inches INT;
-> SET yards=TRUNCATE(p_inches/36,0);
-> SET feet=TRUNCATE((p_inches%36)/12,0);
-> SET inches=p_inches%12;
-> SELECT yards,feet,inches;
-> END;
-> $$
ERROR 1304 (42000): PROCEDURE sp_convert_inches already exists
KD1_Rosh_98811> DELIMITER ;
KD1_Rosh_98811> CALL sp_convert_inches(124);
+-----+-----+-----+
| yards | feet | inches |
+-----+-----+-----+
|      3 |     1 |        4 |
+-----+-----+-----+
1 row in set (0.015 sec)

Query OK, 0 rows affected (0.028 sec)
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
KD1_Rosh_98811> DELIMITER $$
KD1_Rosh_98811> CREATE PROCEDURE sp_product(p_num1 DOUBLE,p_num2 DOUBLE)
-> BEGIN
-> IF p_num1*p_num2>=100 THEN
-> SELECT 'Product is equal to or greater than 100' AS result;
-> ELSE
-> SELECT 'Product is less than 100' AS result;
-> END IF;
-> END;
-> $$
ERROR 1304 (42000): PROCEDURE sp_product already exists
KD1_Rosh_98811> DELIMITER ;
KD1_Rosh_98811> CALL sp_product(10,10);
+-----+
| result |
+-----+
| Product is equal to or greater than 100 |
+-----+
1 row in set (0.003 sec)

Query OK, 0 rows affected (0.026 sec)
```